Analysis of the Effects of Land Toxicity and Crop Diversity on Winter Mortality

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July 2026, revised October 2026

1. Methodology

The spatial model calculates the proportion of various land use types within a specific radial distance from the apiary. It then applies a toxicity weight based on the aggregated volume of chemicals per hectare. A foraging factor of 0/1 is assigned to each land use type to reflect whether honeybees forage for nectar within that specific land category. Both indices were calculated at two spatial scales: a radial buffer of 500 meters and 1000 meters, which equal roughly 78 and 314 hectares of land.

Land-use data could be linked to 423 of the 828 apiaries in the cleaned survey. The analysis in this report is limited to those apiaries.

1.1. Toxicity Index

1) **Foraging Factor:** This parameter attempted to quantify the likelihood that a honeybee would actively interact with a specific land class. In this preliminary iteration, it was conceptualized strictly as a binary switch. A value of 1 was assigned to suspected agricultural and floral lands, while a value of 0 was assigned to lands deemed non-forageable, such as fresh water bodies, deep coniferous forests, or heavily paved primary infrastructure. This binary approach sought to exclude spatial areas that theoretically contribute no oral or contact chemical exposure.
2) **Pesticide Factor:** This parameter aimed to represent the intensity of chemical application inherent to specific agricultural or land-management practices. Drawing loosely on average aggregate pesticide usage data published by the Centraal Bureau voor de Statistiek (CBS) in kilograms per hectare, the factor was assigned relative to high-input commercial crops. For example, maize (corn), known for heavy herbicide application and the widespread use of systemic insecticidal seed coatings, was assigned a baseline index value of 1.0. Other crops, such as intensively sprayed commercial fruit orchards or flower bulb fields, were scaled mathematically relative to this maize benchmark.

Calculation formula

For each apiary and buffer radius, the toxicity index is calculated as:
Toxicity Index = sum(foraging factor x pesticide factor x oppervlakte_procent)

$$Toxicity\ Index = \sum(Foraging\ Factor_i \times\ Pesticide\ Factor_i \times\ Oppervlakte\ Procent_i)$$

1.2. Crop Diversity Index

Calculation formula

For each apiary and buffer radius, the diversity index is calculated as:

Crop Diversity Index = count(land types where pesticide factor > 0 and oppervlakte_procent > 1%)
Only land types with a pesticide factor above 0 are counted. These are mainly agricultural crops such as maize, potatoes, fruit and flower bulbs. A type counts once when it covers more than 1% of the buffer, whatever its size, so grassland, forest, water and roads never count. The index is a whole number: 0 means no crop type covers more than 1% of the buffer.

$$Diversity\ Index = \sum \text{Eligible Land Type}_i$$

1.3. Input and Output Files

Input files

output_2025_dot.csv
This document is the distribution of land-use types within the 500-meter and 1,000-meter radius around each apiary. The land area is in: a. hectares (oppervlakte_ha), b. as a proportional share of the buffer area (oppervlakte_procent).

land_types.csv
This document contains the assigned foraging factor and pesticide factor for each land-use type. These factors are used to estimate the relative toxicity contribution of each land category.

Output Files

04_toxicity_location_index_oppervlakte_procent.csv
The calculated toxicity index for each apiary within a 1000 meter and 500 meter radius, using oppervlakte_procent as the weight.

04_crop_diversity_location_index.csv
The number of crop types covering more than 1% of the buffer, for each apiary within a 1000 meter and 500 meter radius.
Both output files are written by the notebook 04_landscape_indices.ipynb in the project's GitHub repository; the models below are in 05_gee_toxicity_diversity.ipynb.

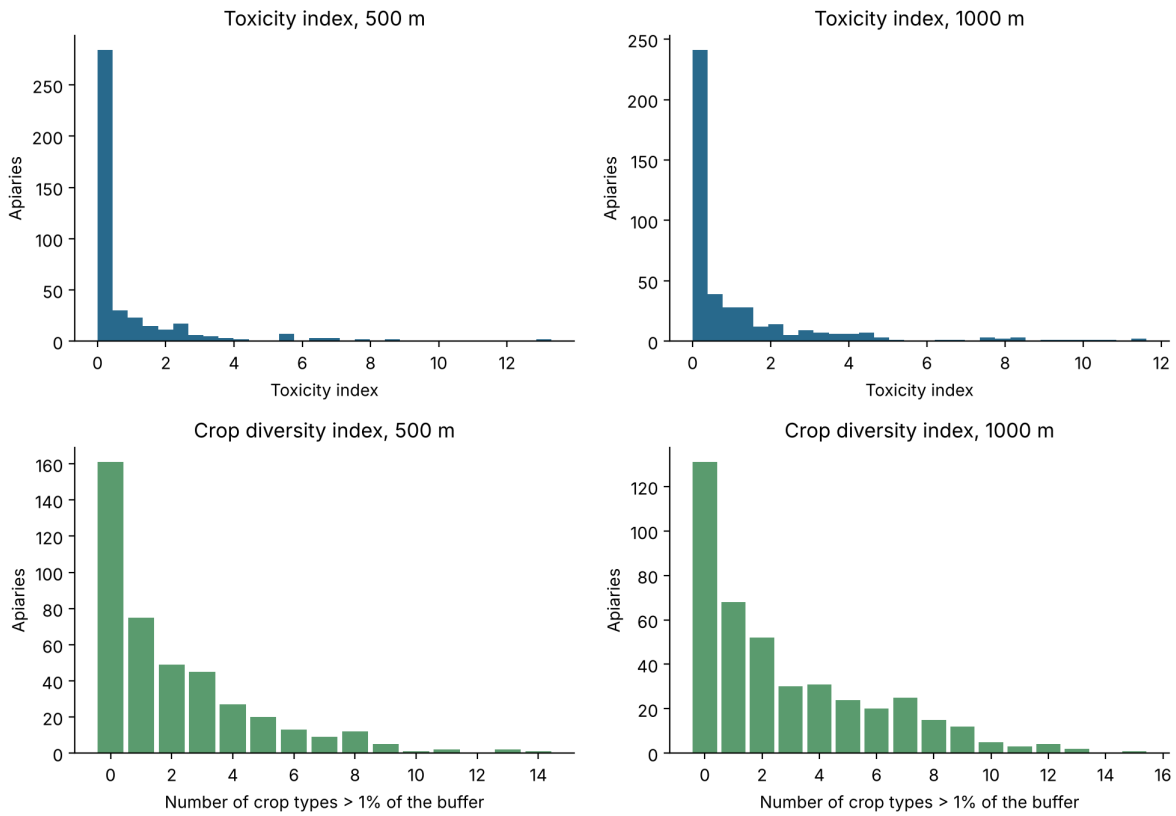


Figure 1. Distribution of the toxicity index and the crop diversity index across apiaries.

1.4. The Generalized Estimating Equations (GEE) Framework

A fundamental methodological challenge in apiary research is that individual hives or multiple apiary sites managed by the same beekeeper are not statistically independent entities. A single operator generally applies highly uniform feeding, treating, and physical handling practices across all their colonies. Applying standard Generalized Linear Models (GLMs) or simple Ordinary Least Squares (OLS) regressions to such data violates the core statistical assumption of independence, leading to overly confident p-values.

To rectify this, the analysis deployed Generalized Estimating Equations (GEE). The GEE is a population-averaged model that inherently accounts for intra-cluster correlation. In this specific framework, all apiaries belonging to the same beekeeper were treated as a discrete mathematical cluster.

The model was strictly specified with a binomial family and a logit link function to handle the proportional outcome variable (death_rate), mathematically defined as the difference between colonies entering the winter (2025_winter_in) and surviving the winter (2026_winter_uit), divided by number of colonies entering winter.

Each apiary is weighted by its number of colonies entering winter, so every colony counts once. The working correlation is independence and the standard errors are robust (sandwich) estimates. These are the same settings as the main mortality models in the key-conclusions

report.
Sample: the full-sample models use 418 to 420 apiaries from 329 to 330 beekeepers (about 1,990 colonies). The honey-only models use 70 to 72 apiaries from 35 to 36 beekeepers (about 240 colonies).

1.5. Model Equations and Adjustments

The regression analysis was conducted in two tiers for both the toxicity and crop diversity analysis:

1. **Unadjusted Model:** This simple regression model only includes the primary independent variable of interest (either the Toxicity Index or the Crop Diversity Index). It is designed to assess the standalone mathematical relationship between the environmental index and the winter death rate, serving as a baseline.
2. **Adjusted Model:** This comprehensive model incorporates the independent variable alongside a specific set of categorical control variables, i.e. the beekeeping method (imker_methode), winter feeding regimen (winter_food), and Varroa mite treatment protocol (varroa_treatment). Adding these controls reduces the risk that an environmental effect is really a difference in beekeeping practice. The reference groups are the controlled beekeeping method, honey + sugar feeding and the alternative full varroa treatment.

2. Results: Full Sample

2.1. Toxicity Index

Empirical Hypothesis

The core biological expectation driving this research was predicated on the widely held consensus that higher exposure to chemically affected landscapes has a negative effect on insect health. Therefore, we propose the null hypothesis and the alternative hypothesis as follows:

- **H0**: There is no positive correlation between the toxicity index and the winter death rate.
- **H1**: There is a positive correlation between the toxicity index and the winter death rate.

Results

The continuous toxicity_index variable was included into the GEE framework alongside the categorical control variables (imker_methode, winter_food, and varroa_treatment). The models were run iteratively for both the 500-meter and 1000-meter spatial radius, under both unadjusted (evaluating toxicity alone) and adjusted (adding the control variables) parameters. The output from the GEE binomial regressions revealed counter-intuitive results that challenge the initial research assumption.

The critical regression metrics evaluating the toxicity index are synthesized in the table below:

Regression Metric	500m Unadjusted GEE Model	1000m Unadjusted GEE Model	500m Adjusted GEE Model	1000m Adjusted GEE Model
Coefficient (Toxicity Index)	-0.028	-0.043	-0.071	-0.091
Robust Standard Error	0.048	0.054	0.033	0.038
p-value	0.565	0.422	0.031 (*)	0.016 (*)
Odds Ratio (per 1 unit increase)	0.973	0.958	0.931	0.913
95% Confidence Interval	[0.885, 1.069]	[0.862, 1.064]	[0.873, 0.994]	[0.849, 0.983]

The statistical results showed that the analysis fails to reject the null hypothesis (H0). In the adjusted models, the data actually indicate a statistically significant negative association ($p = 0.031$ at 500 m and $p = 0.016$ at 1000 m). Without the control variables the association is weaker and not significant ($p = 0.56$ and 0.42).
According to the adjusted 500-meter model, a 1-unit increase in the toxicity index is associated with about 6.9% lower odds of colony loss. In the 1000-meter model this is 8.7%. In other words, after accounting for beekeeping practices, apiaries in areas with a higher toxicity index lost slightly fewer colonies than apiaries in low-toxicity areas. This is an association; it does not show that pesticides are harmless to bees.

We propose two primary explanations for this result:
1. On the methodological level, it is possible not all pesticides have a negative impact on honeybee health or accumulate in pollen, nectar, and ultimately honey. The current toxicity index takes a broad spectrum of pesticides into consideration, it might include many benign

substances. As a result, the impact of those specific toxic compounds may be masked by the less related ones.
2. From an ecological perspective, heavy agricultural zones, despite larger quantities of pesticides, may provide highly concentrated flowers during spring and summer. This allows the bees to have enough nutrients to withstand the toxicological and parasitic threats better than those undernourished bees in structurally diverse but nutritionally deficient natural reserves.

2.2. Crop Diversity Index

Empirical Hypothesis

The core biological expectation driving this research was predicated on the widely held consensus that greater crop diversity has a positive effect on insect health.

- **H0**: There is no negative association between the crop diversity index and the winter death rate.
- **H1**: A higher crop diversity index is associated with a lower winter death rate.

Results

The continuous crops_diversity_index variable was included into the GEE framework alongside the categorical control variables (imker_methode, winter_food, and varroa_treatment). The models were run iteratively for both the 500-meter and 1000-meter spatial radius, under both unadjusted (evaluating crop diversity alone) and adjusted (adding the control variables) parameters.
The output from the GEE binomial model supports the initial research assumptions.

The critical regression metrics evaluating the crop diversity index are synthesized in the table below:

Regression Metric	**500m Unadjusted GEE Model**	**1000m Unadjusted GEE Model**	**500m Adjusted GEE Model**	**1000m Adjusted GEE Model**
Coefficient (Crop Diversity Index)	-0.082	-0.086	-0.099	-0.094
Robust Standard Error	0.059	0.040	0.041	0.029

p-value	0.169	0.029 (*)	0.015 (*)	0.001 (**)
Odds Ratio (per 1 unit increase)	0.922	0.917	0.906	0.910
95% Confidence Interval	[0.820, 1.035]	[0.849, 0.991]	[0.836, 0.981]	[0.860, 0.964]

At both radii, the adjusted models show a statistically significant negative association between the crop diversity index and the winter death rate (p = 0.015 at 500 m and p = 0.001 at 1000 m); the unadjusted 1000 m model is also significant (p = 0.029). The analysis therefore rejects H0 in favour of H1. Each additional crop type covering more than 1% of the buffer is associated with about 9% lower odds of winter colony loss (odds ratio 0.906 at 500 m and 0.910 at 1000 m).

Thus, apiaries surrounded by a greater variety of crops survived the winter better. This is an association: crop diversity may also reflect other differences between locations.

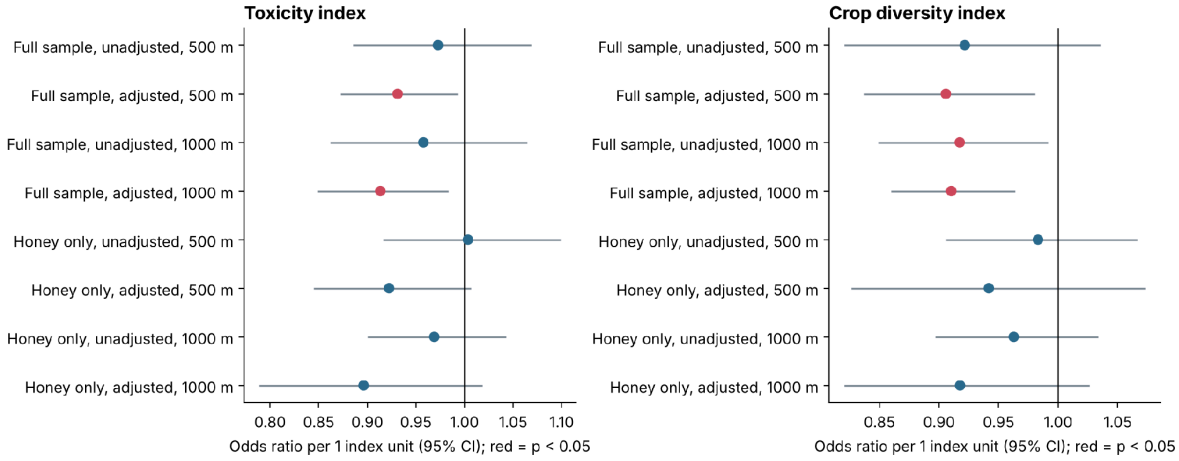


Figure 2. Odds ratio per 1 unit of each index, for every model in sections 2 and 3 (95% confidence interval; red = p < 0.05). Below 1 means fewer colony losses.

3. Results: Subgroup of Bees only Fed on Honey

There lies the assumption that the bees only fed on honey are more sensitive to environmental effects than bees with more artificial feeding. Therefore, this subgroup isolates colonies strictly

based on winter feeding, evaluating exclusively those beekeepers who fed their colonies entirely their own natural honey during the winter.

This subgroup is small (70 to 72 apiaries from 35 to 36 beekeepers), so the estimates are imprecise. In the adjusted models some beekeeping-method groups have almost no losses, which makes those control coefficients unstable; the index coefficients are still estimated. Winter feeding is constant in this subgroup and is left out of the adjusted models.

3.1. Toxicity Index

Empirical Hypothesis

Similarly, we propose the null hypothesis and the alternative hypothesis as follows:

- **H0**: There is no positive correlation between the toxicity index and the winter death rate.
- **H1**: There is a positive correlation between the toxicity index and the winter death rate.

Results

The critical regression metrics evaluating the toxicity index are synthesized in the table below:

Regression Metric	500m Unadjusted GEE Model	1000m Unadjusted GEE Model	500m Adjusted GEE Model	1000m Adjusted GEE Model
Coefficient (Toxicity Index)	0.004	-0.032	-0.081	-0.110
Robust Standard Error	0.046	0.038	0.045	0.065
p-value	0.938	0.401	0.071	0.092
Odds Ratio (per 1 unit increase)	1.004	0.969	0.922	0.896
95% Confidence	[0.917, 1.099]	[0.900, 1.043]	[0.845, 1.007]	[0.789, 1.018]

Interval				

The analysis fails to reject the null hypothesis (H0). There is no evidence of a positive association between the toxicity index and mortality in the honey-only subgroup. The adjusted coefficients are negative, as in the full sample, but not significant at the 5% level (p = 0.071 at 500 m and p = 0.092 at 1000 m).

3.2 Crop Diversity Index

Empirical Hypothesis

Similarly, we propose the null hypothesis and the alternative hypothesis as follows:

- **H0**: There is no negative association between the crop diversity index and the winter death rate.
- **H1**: A higher crop diversity index is associated with a lower winter death rate.

Results

The critical regression metrics evaluating the crop diversity index are synthesized in the table below:

Regression Metric	500m Unadjusted GEE Model	1000m Unadjusted GEE Model	500m Adjusted GEE Model	1000m Adjusted GEE Model
Coefficient (Crop Diversity Index)	-0.017	-0.038	-0.060	-0.086
Robust Standard Error	0.042	0.036	0.067	0.057
p-value	0.682	0.296	0.369	0.132
Odds Ratio (per 1 unit	0.983	0.963	0.942	0.918

increase)				
95% Confidence Interval	[0.906, 1.067]	[0.897, 1.034]	[0.826, 1.073]	[0.821, 1.026]

The statistical analysis ***failed to reject the null hypothesis (H_0).*** Consequently, the results do not show a statistically significant effect of crop diversity on colony winter survival within the subgroup fed exclusively on honey during winter.

4. Exploratory Analysis: Feeding and Varroa Treatment

The July 2026 analysis also explored whether the toxicity association depends on winter feeding or on varroa treatment. These models were not planned in advance and test several things at once, so they should be read as ideas for further work, not as findings.

(a) Strict honey-only versus strict sugar-only feeding (mixed feeding left out). (b) The toxicity slope by varroa-treatment group: full 3-step treatment (reference), partial or alternative treatment, and no treatment. Both models control for the beekeeping method, and (a) also for varroa treatment, (b) for winter feeding.

The table shows odds ratios per 1 unit of the toxicity index, with p-values in brackets.

Toxicity index term	**500 m**	**1000 m**
Toxicity slope, sugar only	1.008 (p = 0.926)	0.815 (p = 0.317)
Toxicity slope, honey only	0.937 (p = 0.107)	0.907 (p = 0.075)
Difference honey vs sugar (interaction)	0.956 (p = 0.639)	1.112 (p = 0.577)
Toxicity slope, full varroa treatment	0.794 (p = 0.048)	0.728 (p = 0.020)
Difference partial/alternative vs full (interaction)	1.219 (p = 0.120)	1.270 (p = 0.119)

Difference no treatment vs full (interaction)	1.228 (p = 0.106)	1.329 (p = 0.047)

Feeding: the toxicity slope does not differ clearly between honey-only and sugar-only apiaries (interaction p = 0.64 at 500 m and 0.58 at 1000 m).

Varroa treatment: the negative toxicity association comes mainly from apiaries with full 3-step treatment (odds ratio 0.79 at 500 m, 0.73 at 1000 m). In untreated apiaries the slope is close to flat (odds ratio 0.97 and 0.97); the difference is significant at 1000 m only (p = 0.047). One possible reading is that well-treated colonies in agricultural areas benefit from the extra forage, while untreated colonies do not. With this many tests and no correction for multiple testing, this needs confirmation in next year's data.

Appendix

Link to the survey:

https://app.formbricks.com/environments/cmk4775is8p8ead01d6uk33ow/surveys/cmkbibhk69hohad01t13iqs8d/summary?referer=true
https://docs.google.com/document/d/19fNeGK6ufE2Ipf6e1StW3Igc7iy2z50FV9FWAzZc4UU/edit?tab=t.0#heading=h.2yjxr1umsg35

Deleted data entries:

Location not in the Netherlands: 3 entries
Abnormal growth during winter: 9 entries
0 colonies going into winter: 4 entries

Data cleansing and coding:

Words in bold are original options

Q4. Welke imkermethode hanteer je?
1: **Aalster methode**; AANGEPASTE AALSTER; **Renson methode**; Darrenraatmethode; KI; Breek zwermcellen; VSH focus BeeBreed; Arrest methode
2: **Biodynamisch imkeren**; **Korfimkeren**; Natuurlijk imkeren; Natuurvriendelijk imkeren; Darwiniaans imkeren; Biologisch met SKAL certificaat; Topkasten; Ongeregelde bijenkorf; Zwermen afwachten; Niet volledig BD, maar sterke invloeden hieruit
3: **Ik imker regulier**; **Eenbaks imkeren**; Liebig; Methode Aumeier; Golzkasten; op een broedkamer in warmbouw; Golz; Broedaflegger

4: **Weet ik niet**; **Ik gebruik geen specifieke methode**; +mixed methods; Natuurbouwkast en gangbare spaarkast. beide. Zwermen en geen behandeling varia; Topkasten met arrestraammethode; In de verenigingshal doen de leden ieder hun eigen ding; Geleerd van imkerij Drakenbijen;
Dummy Coding

Q19. Hoe was de ontwikkeling van de volken tijdens het voorjaar en zomer 2025 op deze bijenstand?
1: **Overwegend goed**; Ik had geen dode volken; heel goed; uitzonderlijk goed
2: **Overwegend matig**; Langzaam
3: **Overwegend slecht**
4: Nvt

Q20. Hoe heb je de bijen ingewinterd?
1: **Uitsluitend op suikerwater(bijenvoer)**; suikerwater (1 deel water, 2 delen suiker) met kruidenthee (kamille, heermoes, duizendblad).
2: **Combinatie van eigen honing en suikerwater(bijenvoer)**; Combinatie eigen honing met invertsuiker; Eigen honing (ik slinger niet), aangevuld met suiker vanwege drachtarmoede op platteland; Eigen honing in kast aangevuld met kruidentheesuierwater; Een van de combinaties hierboven en extra eiwitten; Eigen honing en BeeElixer aangevuld met kruiden zoals kamille en goudsbloem; Fondant toegevoegd op 30-1-26. Ik heb daar nu spijt van; 3 volken met suiker ingewinterd, rest had in de Biesbosch voldoende voer in de onderbak. Op suiker en honing maakt geen verschil bijna alles goed in het voorjaar; Eigen honing, suikerwater met etherische oliën.
3: **Uitsluitend op eigen honing, Combinatie van eigen honing en honing van andere volken**; was voldoende zowiezo geen honing afgenomen; Ik geef geen voer. Ik neem ook geen honing af. Ik imker met korven; Voor 90% op eigen honing; Met suikerwater maar ben gestopt toen ik ontdekte dat er nog erg laat (tot eind september) dracht binnen kwam
Ordinal (1-lower death)

Q21. Heb je het afgelopen jaar varroa waargenomen bij je eigen volken op deze bijenstand ?
1: Nee, geen varroa waargenomen
2: Ja, weinig
Uiteraard nog last van mijten, maar onze teelt is volledig gericht op VSH met o.a. KI, waardoor bij 60% van de volken de mijtentelling uit kwam onder 1%
alleen op de varroa lade
Een volk behandeld met mierenzuur en in de herfst alles gesubsidieerd
3: Ja, matig
sommige volgen veel, sommige volken weinig
Ja, maar hoeveel? geen idee.
Sommige volken wel, andere niet

4: Ja, veel
5: Weet ik niet, ik heb er niet op gelet
ik doe mee aan het WUR project, Weerbare bijen - Darwin Methode

Q22. Zijn de bijen het afgelopen jaar tegen varroamijt behandeld op deze bijenstand?
1: **Ja, volledige behandeling met 3 gangenmenu**
2: **Ja, alternatieve behandeling**
3: **Ja, gedeeltelijke behandeling alleen in de winter**
4: **Ja, gedeeltelijke behandeling alleen in de zomer**
5: **Nee**
Dummy Coding

Q23. Heb je het afgelopen jaar Aziatische hoornaars waargenomen bij je volken op deze bijenstand ?
1: **Nee, geen Aziatische hoornaar waargenomen**
Veel wespen, hommelnest, Hoornaar gezien (kon niet zien of deze Europees of Aziatisch was.)
Wel veel Europese Hoornaars die bijen pakten
niet direct bij het volg zelf AG gezien, maar er waren waarnemingen in de omgeving
Dit jaar voor het eerst veel last van actief prederende Europese hoornaars
2: **Ja weinig, af en toe 1**
We hebben er 1 gezien, die voor de kast heen en weer vloog. wij stonden te kijken wat erzou gebeuren en ineens vloog hij met een zwaai door de lucht. Hij werd weggevoerd door 3 bijen al vechtend. Zeker 1 bij liet hierbij het leven. De hoornaar hebben we niet weergezien.
1 dode hoornaar
In het voorjaar 1 koningin en gevangen.
Wel op 200 mtr afstand een beginnend nest verwijderd
Zat wel een nest op 100 meter van waar mijn kast staat
Veel Europese hoornaars,en eenenkele Aziaat
in april heb ik 1 AH koningin gevangen en doorgegeven op waarneming.nl
Bij de kast een enkele gezien/ gevangen. Echter de bijen bleven bij de kast hangen en gingen er mid zomer er niet op uit, ookal was het 25 graden. Later hebben we een nest gevonden 500 meter verder op. Ik vermoed dat de bijen in de open lucht er wel laat van hadden en daarom besloten thuis te blijven.
3: **Ja matig, vaak minstens 1 voor de kast**
4: **Ja veel, er zijn er vaak meer dan 3 voor de kast te zien**
5: **Weet ik niet**
ik begin net we gaan het dit jaar zien
Dummy Coding

Q24. Heb je het afgelopen jaar deformed wings virus waargenomen bij je volken op deze bijenstand?
1: **Nee, geen deformed wings waargenomen**
2: **Ja, weinig**
3: **Ja, matig**
4: **Ja, veel**
5: **Weet ik niet**